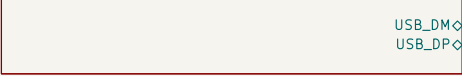


WINDSCREEN POD BOARD: block diagram. Each box is a sheet (double-click it in KiCad).
Power rails +3V3, VBUS and GND join the sheets by name (power symbols);
every other connection between sheets is a wire here.

USB-C to the laptop



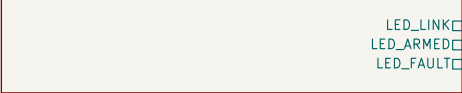
File: usb.kicad_sch

Sensors and local connectors



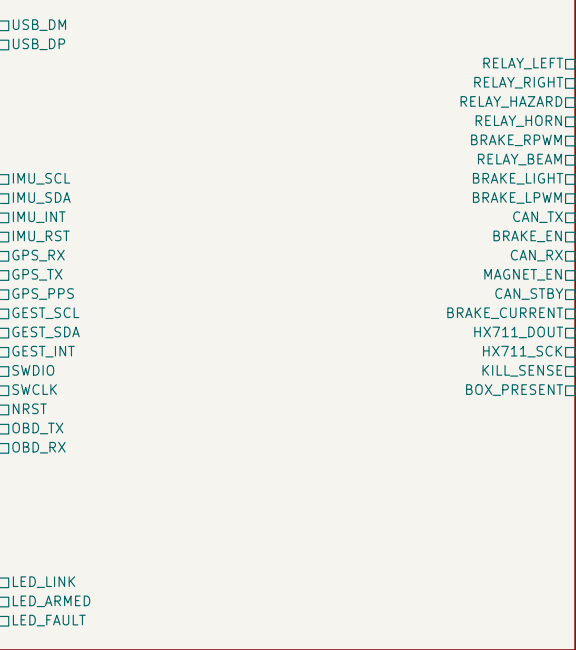
File: sensors.kicad_sch

Status LEDs (driver-facing edge)



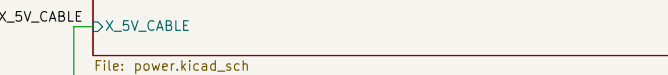
File: leds.kicad_sch

MCU: STM32F405RGT6

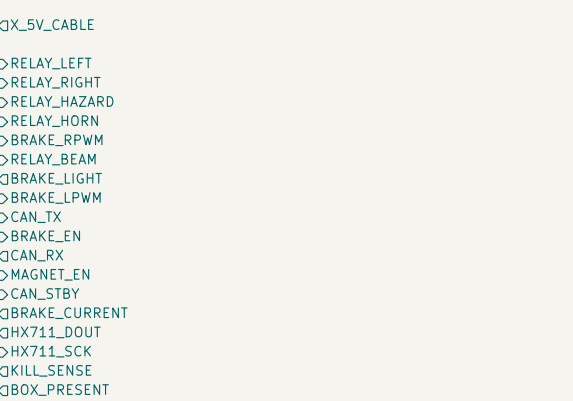


File: mcu.kicad_sch

Pod supply: USB or cable 5 V, 3.3 V regulator



Inter-box cable: DB-25, series resistors, input filters



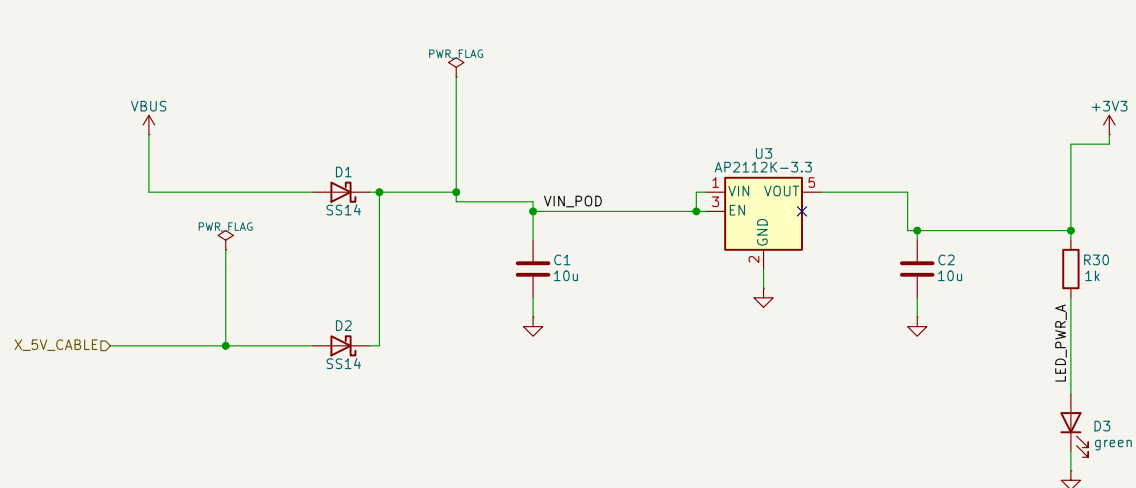
File: cable.kicad_sch

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)
AR Windscreen FYP – Klogora Ian Mwenda
Sheet: /
File: pod_board.kicad_sch

Title: Windscreen pod board (MCU, sensors, USB, cable)

Size: A3	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 1/7

POD SUPPLY: two 5 V sources, diode-OR'd: the laptop's USB (D1) and the power box through the cable (D2). Either one runs the pod, so it can be programmed and tested on USB alone. U3 makes 3.3 V for the MCU and sensors. D3: 3.3 V present.



Mounting holes: M3, no electrical connection

H1
M3

H2
M3

H3
M3

H4
M3

Generated by hardware/gen; see BUILD_GUIDE Part 4.8

Windscreen pod board (MCU, sensors, USB, cable)

AR Windscreen FYP – Kiogora Ian Mwenda

Sheet: /Pod supply: USB or cable 5 V, 3.3 V regulator/

File: power.kicad_sch

Title: Pod supply: USB or cable 5 V, 3.3 V regulator

Size: A4

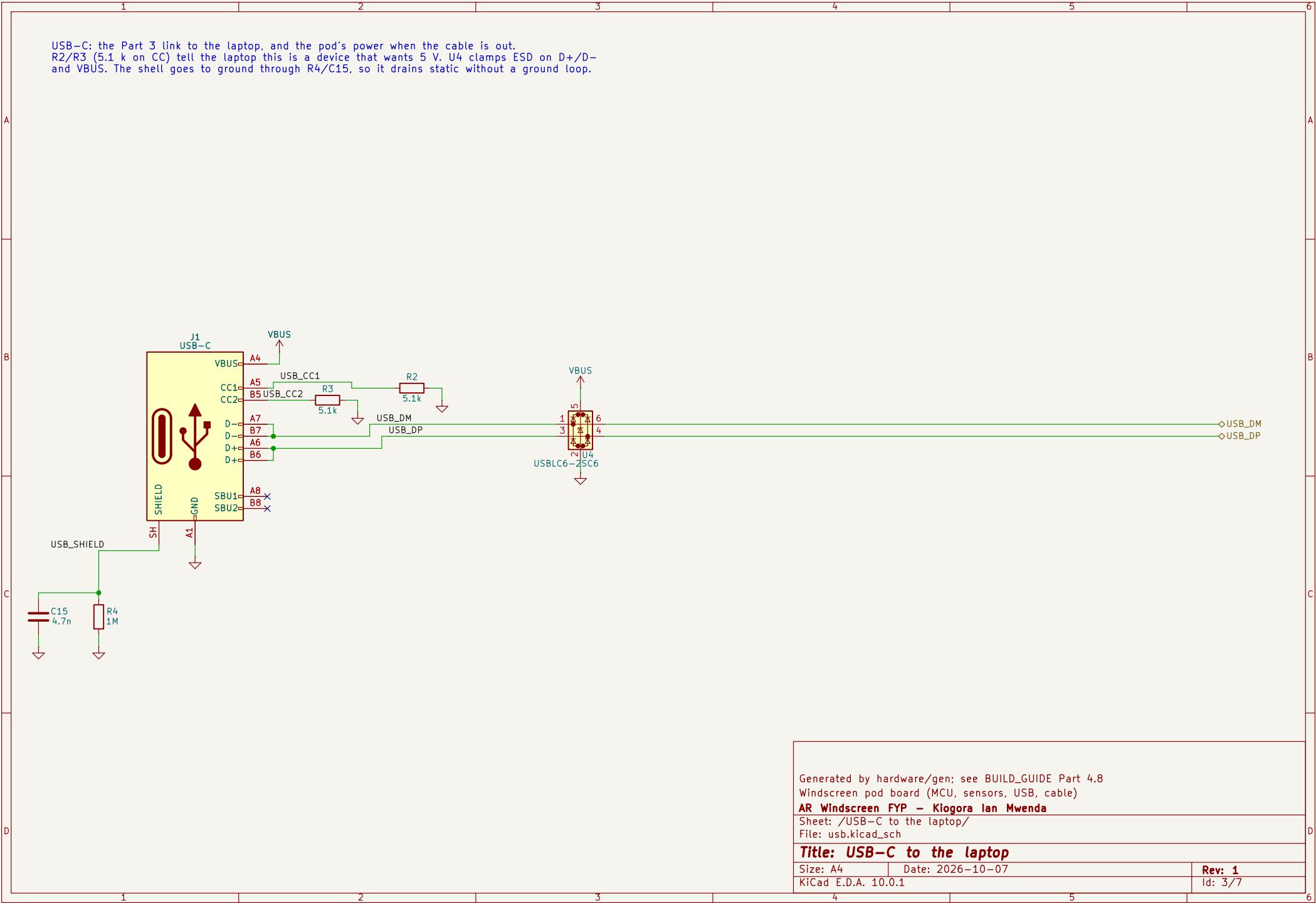
Date: 2026-10-07

Rev: 1

KiCad E.D.A. 10.0.1

Id: 2/7

USB-C: the Part 3 link to the laptop, and the pod's power when the cable is out.
R2/R3 (5.1 k on CC) tell the laptop this is a device that wants 5 V. U4 clamps ESD on D+/D- and VBUS. The shell goes to ground through R4/C15, so it drains static without a ground loop.



Generated by hardware/gen; see BUILD_GUIDE Part 4.8

Windscreen pod board (MCU, sensors, USB, cable)

AR Windscreen FYP – Kiogora Ian Mwenda

Sheet: /USB-C to the laptop/

File: usb.kicad_sch

Title: USB-C to the laptop

Size: A4

Date: 2026-10-07

Rev: 1

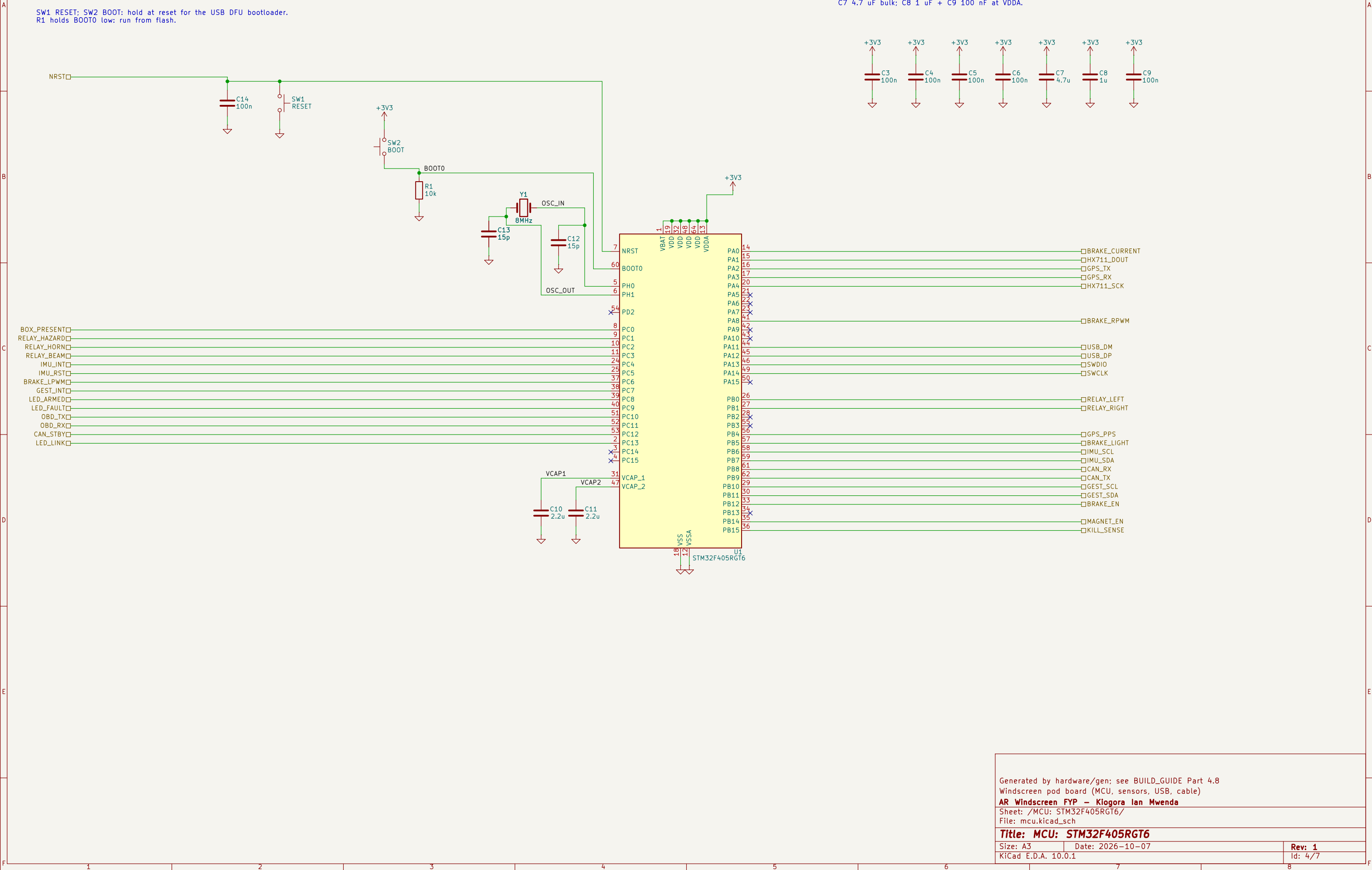
KiCad E.D.A. 10.0.1

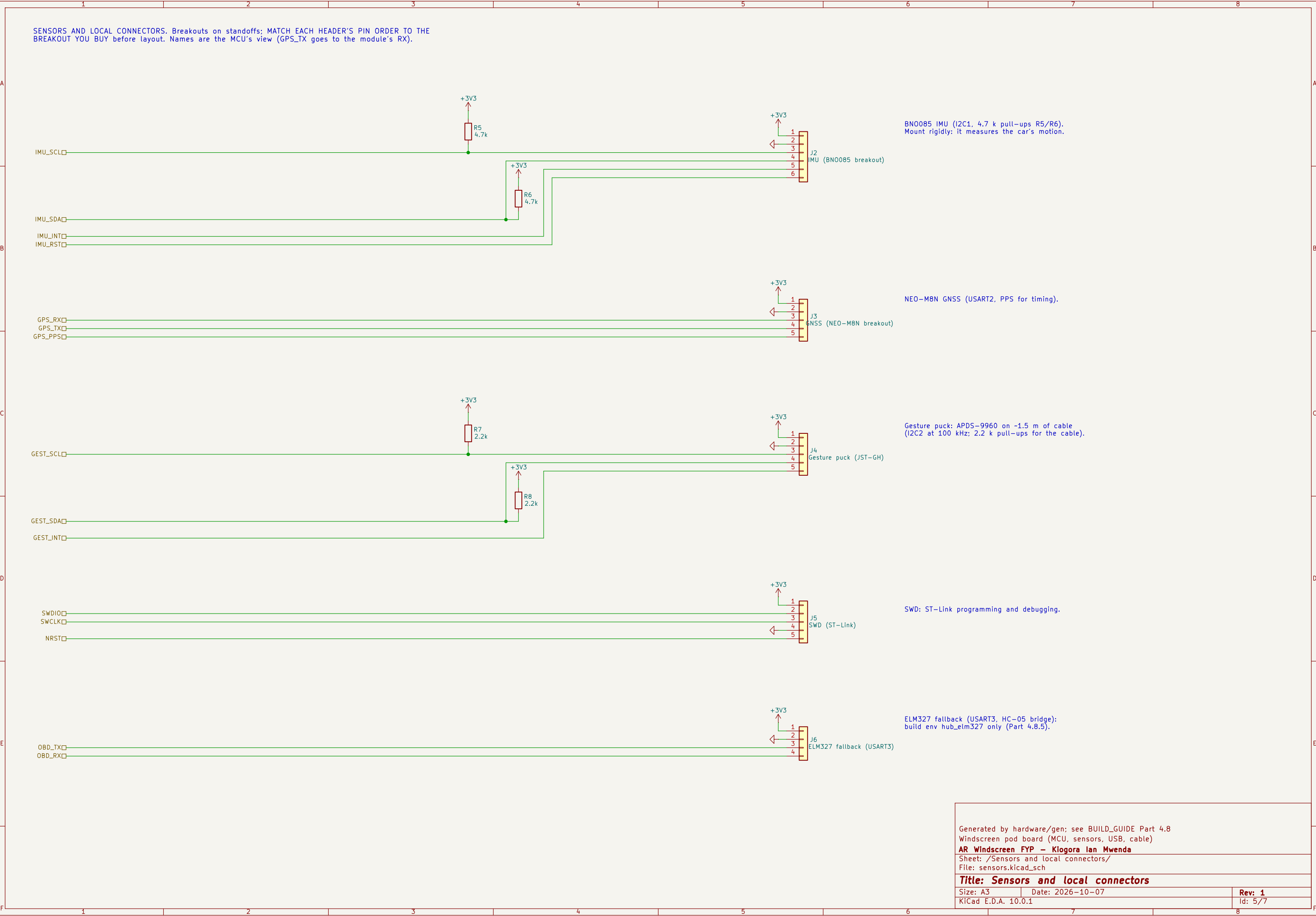
Id: 3/7

MCU. Every pin assignment equals firmware Config.h (checked by check_nets.py). Each pin runs straight out to the sheet that uses it. Left: reset, boot, the 8 MHz crystal, the core regulator's capacitors (VCAP), and port C. Right: ports A and B.

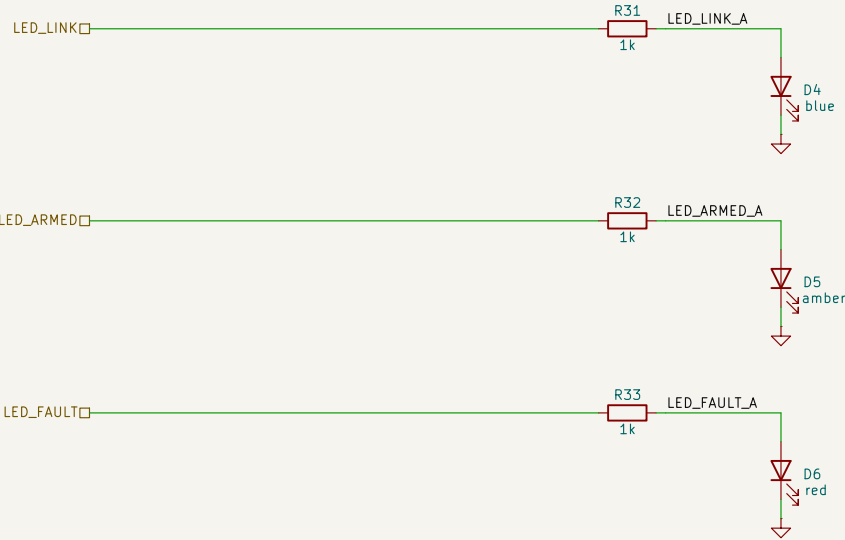
SW1 RESET; SW2 BOOT: hold at reset for the USB DFU bootloader.
R1 holds BOOT0 low: run from flash.

Decoupling: C3–C6 100 nF, one beside each VDD pin;
C7 4.7 uF bulk; C8 1 uF + C9 100 nF at VDDA.





STATUS LEDs on the board's top edge, facing the driver: LINK (blue) the laptop is talking, ARMED (amber) the hub accepts brake commands, FAULT (red) a latched fault (Part 4.5).



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Windscreen pod board (MCU, sensors, USB, cable)

AR Windscreen FYP – Kiogora Ian Mwenda

Sheet: /Status LEDs (driver-facing edge)/
File: leds.kicad_sch

Title: Status LEDs (driver-facing edge)

Size: A4	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 6/7

